Reading list — GATEKEEPER

Scope before Reuse: Risk-Controlled Admission of Learned Coding Skills

FSE 2027 Research Track · `FSE-2027-13` · proposal P14 (rank 13)

A skill can be fully authorized, pass every replay test, and still be the wrong procedure for this repository.

Read in this order

If you read nothing else, read these, in this sequence. Each line says what you should be able to answer afterwards.

1. **GSE** [9] — read first and read closely; the decisive baseline, and the reason the headline margin is 6.4 rather than 17.6 points.
2. **Formal skill verification** [7] — capability containment, then articulate why authorized is not safe.
3. **Voyager** [18] — the skill-library pattern in its original form.
4. **Angelopoulos et al.** [1] — risk control by calibration; the statistical frame.
5. **Betting confidence sequences** [20] — admission at a data-dependent trial count.
6. **Conformal risk control** [2] — distribution-free guarantees and the exchangeability that transfer violates.

Closest prior work — the papers this must be distinguished from

These occupy the neighbouring territory. The contribution is only defensible relative to them, and a reviewer will know them.

GSE [9]
Globally reusable coding skills with inter-skill relations, consolidation and replay verification. The decisive baseline; distinguishing from it is the paper's central obligation.

Formal skill verification [7]
Declared capability containment. Correct, and answering a different predicate from functional appropriateness.

MemCoder [12]
Structured intent-to-code experience reuse.

MemGovern [13]
Governed development histories as experience cards.

Voyager [18]
The skill-library pattern; where reuse-without-scope originates.

Foundations — the machinery being used

Claimed as building blocks, not contributions. Know them well enough to state precisely what is standard and what is not.

Learn Then Test [1]
Calibrating a procedure to achieve risk control over a declared family.

Conformal risk control [2]
Distribution-free risk control under exchangeability.

Algorithmic learning in a random world [17]
Conformal prediction foundations.

Betting confidence sequences [20]
Anytime-valid admission decisions.

Time-uniform confidence sequences [10]
Validity at data-dependent stopping.

Benjamini–Hochberg [3]
Simultaneous control over a scope family.

Program slicing [21]
Interaction footprints are slices.

Benchmarks, baselines and evaluation practice

What the evaluation is built on, compared against, or measured with.

SWE-bench [11]
Target task substrate.

SWE-bench-Live [22]
Cross-repository transfer targets.

OpenHands [19]
Harness.

Cluster bootstrap [4]
Contexts from one source skill are correlated.

CooperBench [6]
Multi-skill interaction settings.

Wider context

Useful for framing and for the related-work section; not on the critical path.

GEM / MemState [8]
Validity of facts; kin to scope conditions, discharged differently.

TMA-NM [16]
Authority over agent memory and skills.

AgentPoison [5]
Poisoned skills; the adversarial-transfer regime's motivation.

ReasoningBank [15]
Reusable strategy distillation.

Automatic software repair survey [14]
What the skills are procedures for.

Citation caveat. Entries drawn from 2026-dated preprints inherit their metadata from the source audit document and are flagged `% VERIFY` in `refs.bib`. Verify each against DBLP / arXiv / DOI before citing it in a submission; see `PLAN.md`, *Citation integrity gate.*

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